GAME THEORY:

APPLIED TO COMPETITON & INDUSTRIAL ORGANIZATION

Session 13

Prof. Ricardo López Pérez

rlp@faculty.ie.edu

Agenda for today

Some Game Theory Extensions:

1. **Repeated Games**
2. **Dynamic vs Static Model**
3. **Mixed Strategies** (probability of the pay-off matrix)
4. **Bertrand model**: imperfect competition only in prices.
5. **Cournot model**: competition in quantities

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Do you remember the Prisoners' dilemma?

		Prisoner A Choices	
		Stay Silent	*Confess and Betray*
Prisoner B Choices	*Stay Silent*	Each serves one month in jail	Prisoner A goes free Prisoner B serves full year in jail
	Confess and Betray	Prisoner A serves full year in jail Prisoner B goes free	Each serves three months in jail

Best response for both players

Nash Equilibrium

The Prisoners' Dilemma

		Prisoner B Choices	
		Stay silent	**Betray**
Prisoner A Choices	**Stay silent**	**(-30, -30)**	**(-365, 0)**
	Betray	**(0, -365)**	**(-90, -90)**

- What happens if you can "repeat" the same decision? **What happens if the Prisoners can make another decision also after observing the outcome of the game? The Nash Equilibrium (NE) still stays the same?**
- **REPEATED GAMES: Iterated Games are simultaneous games repeated infinite (finite) times.**

Iterations (repetitions) of the game change the NE

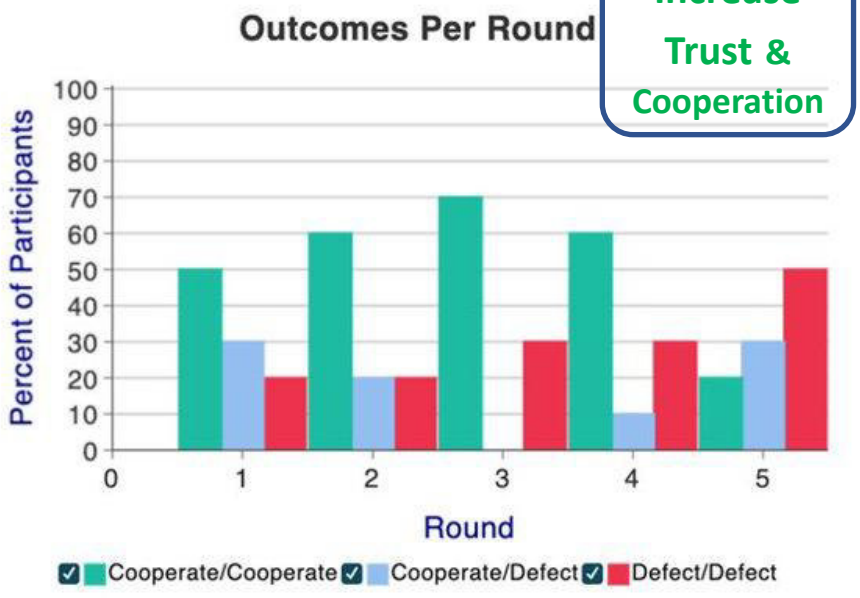
1. Players consider their long-term payoffs instead of short-term gains. This might lead them to behave differently.
2. Repetition creates Sub-games (and subgame perfect Nash equilibrium (SPNE))
3. Repetition creates Learning, Expectations, Trust & Trust Threats
4. Repetitive interactions might create stability of collusive agreement (**COOPERATION**) and Commitment.

The power of reputation to incentivize cooperation

Prisoner's dilemma game with **revealed partners**

Round #	# of Groups	% Choosing C	% Choosing D
1	10	65.00	35.00
2	10	70.00	30.00
3	10	70.00	30.00
4	10	65.00	35.00
5	10	35.00	65.00

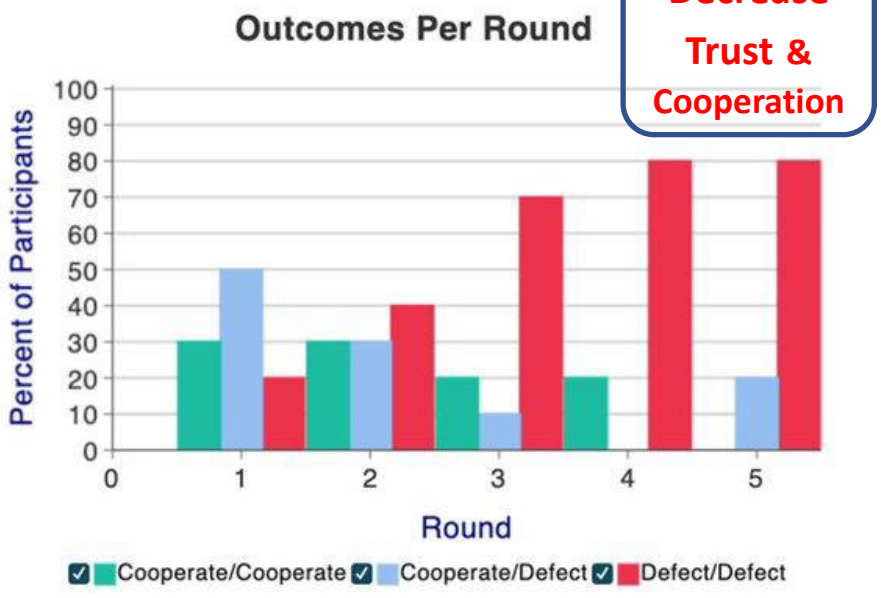
Increase Trust & Cooperation



Prisoner's dilemma game with **anonymous partners**

Round #	# of Groups	% Choosing C	% Choosing D
1	10	55.00	45.00
2	10	45.00	55.00
3	10	25.00	75.00
4	10	20.00	80.00
5	10	10.00	90.00

Decrease Trust & Cooperation



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Dynamic models can address questions that static models cannot answer

(*Example: What determines the intensity of price competition?)*

- **Short-term profits** (in a static model) are often followed by long-term negative effects (in a dynamic model)
- Static models cannot explain how firms can **maintain prices above competitive levels** without formal collusion ("Anti-trust laws")

- Dynamic models try to focus on **COOPERATION** and how price cooperation is sustainable over time.
  - Tit-for-Tat (Spanish →"Dónde las dan las toman")

    Cooperation is a Good strategy on the long run.

    Ex.: Oligopolies in general: Electric Oligopoly; Telecom; Financial institutions,…

    Failure to cooperate damage the industry on the long run. Ex.: Airlines

  - Trigger-grim Strategies

Threatens other players with a "worse," punishment, action if they deviate.

Ex.: The prisoners dilemma, if player A betrays the other, probably player B won´t ever believe player A.

Tit-for-Tat Strategy (Spanish →"Dónde las dan las toman")

- The mere possibility that you play tit-for-tat is sufficient for **competitor to cooperate if the time horizon is long enough**.
- Most managers don't know how long they will be competing with their rivals, serving to make **COOPERATION IS A GOOD STRATEGY**, except near the end (called end game problem). Thus in a repeated game, prisoner's dilemma can have a COOPERATIVE outcome.
- **Industries where only a FEW FIRMS COMPETE under stable demand** cost conditions **may COOPERATE** even though no contractual arrangements are made. *E. g. Electric companies*.
- FAILURE TO COOPERATE is the result of rapidly shifting demand or cost conditions, *e. g. airlines*.



https://www.youtube.com/shorts/QdIUoRuZm9I

Trigger Strategies

In infinitely repeated games we can consider trigger strategies.

A **trigger strategy** essentially **threatens other players with a "worse," punishment, action if they deviate** from an implicitly greed action profile.

- A **non-forgiving** trigger strategy (or **grim trigger strategy**) would involve this **punishment forever** after a single deviation.

Case Santander Bank (1998)

(from Emilio Botín 2011 speech at military Academy of Zaragoza)

- Spain went into the EU in 1986.
- In year 1986 Bank Santander was the 7th commercial Bank in Spain. And they fixed a mission "*Be the 1st commercial bank in the world*"
- At that moment Spanish banks were in a trust.
- 1989 Bank Santander betray the other bankers trust with the "*super account*"
- There were no answer from the other banks in almost 6 moths. After that Bank Santander become one of the most important commercial banks.
- Now → After some mergers and organic growth is on the top 5 biggest commercial bank worldwide.

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- **Pure Strategies** where we assume that all the strategies are chosen with the same probability.

ONE
Nash Equilibrium

Person 2

Person 1	Apply	Don't apply
Apply	(1, 4)	(3, 2)
Don't apply	(2, 3)	(1, 2)

✓ **Unique pure-strategy NE**

(Don't apply, Apply) with payoffs (2, 3)

Neither player can improve their payoff by changing their action alone. *Each player chooses one action with certainty*

- Nash Equilibrium (NE) in **Mixed Strategies** is computed when each player assign a probability to each strategy.

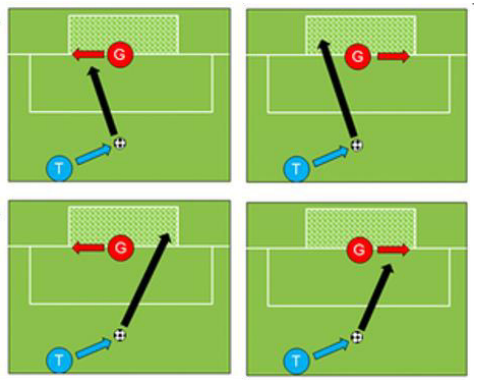
1. **There may be more than one pure-strategy Nash equilibrium**
2. **There may be no pure-strategy equilibrium at all**

When there is no single, clear equilibrium

Soccer penalty shot game

NO
Nash Equilibrium

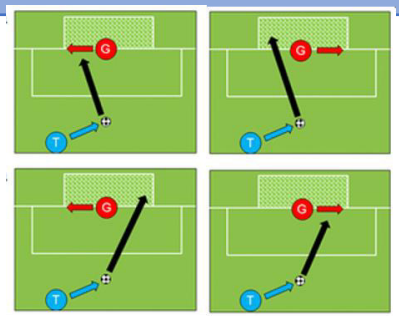
Shooter \ Goalie	Left	Right
Left	(4, 1)	(1, 5)
Right	(1, 2)	(2, 1)



- **Shooter** shoots left or right (from his/her perspective)
- **Goalie** must anticipate and dive left or right (from his/her view) — the goalie cannot wait for the shot and then dive

? Are there any pure-strategy Nash equilibria?

No. Every outcome gives one player an incentive to change



Soccer penalty game

		Goalie	
		Left P(Left) = q	**Right** P(Right) = 1 - q
Shooter	**Left** P(Left) = p	(4, 1)	(1, 5)
	Right P(Right) = 1 - p	(1, 2)	(2, 1)

4q + 1(1 - q) — Shooter's expected payoff from Left

1q + 2(1 - q) — Shooter's expected payoff from Right

1p + 2(1 - p) — Goalie's expected payoff from Left

5p + 1(1 - p) — Goalie's expected payoff from Right

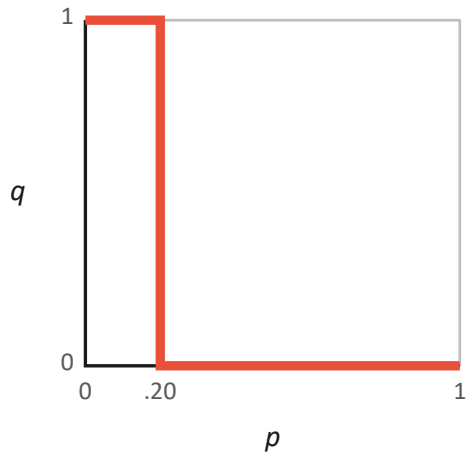
1p + 2(1 - p) = 5p + 1(1 - p)
1p + 2 - 2p = 5p + 1 - 1p
-1p + 2 = 4p + 1
-5p = -1
p = 0.2 (20% shooter chooses left)

4q + 1(1 - q) = 1q + 2(1 - q)
4q + 1 - 1q = 1q + 2 - 2q
3q + 1 = -1q + 2
4q = 1
q = 0.25 (25% goalie chooses left)

Best response functions

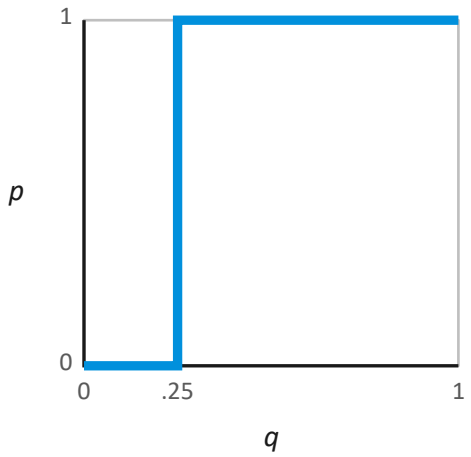
Goalie's best response

Indifferent when p = .20



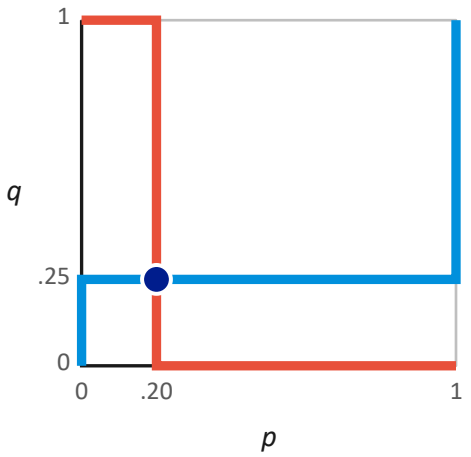
Shooter's best response

Indifferent when q = .25



Mixed-strategy NE

Occurs at p = .20, q = .25



The equilibrium is where each player's strategy is a best response to the other's

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Bertrand Model paradox

- Two firms *1* and *2* compete on the market
- They offer homogeneous (undifferentiated) products. Perfect substitutes.
- Marginal Costs (MC) are the same for both and they are constant.
- Can´t cooperate in any way.
- Demand is a function of the price $D(p_i)$ and is linear.

GAME

- The two firms must set the price simultaneously.
- The optimal price depends on the choice of the other firm (Reaction function).

$$Q_i(p_i) = \begin{cases} Q(p_i) & \text{if} \quad p_i < p_j \\ \alpha_i Q(p_i) & \text{if} \quad p_i = p_j \\ 0 & \text{if} \quad p_i > p_j \end{cases}$$

Because they are perfect substitutes →

- If price of 1 is greater than price 2 then company 1 doesn´t sell anything.

Price 1 > Price 2 → Q(1) = 0

- If price of 2 is greater than price 1 then company 1 doesn´t sell anything.

Price 2 > Price 1 → Q(2) = 0

- ***So, Firm 1 and Firm 2 should figure out the price of the competitor.***
- ***The price should be above the Marginal Costs (MC)***
- ***The equilibrium therefore is:***

$$P_1 = P_2 = MC$$

In a duopoly (with certain conditions) if you compete only in price → the price equilibrium will be only covering the marginal costs

→

ROI = 0 (NO RETURN)

- Bertrand paradox:

- It is possible that in cases of imperfect competition (here, a duopoly), a situation emerges in which firms make zero profits

If Equilibrium occurs → $P_1 = P_2 = MC$

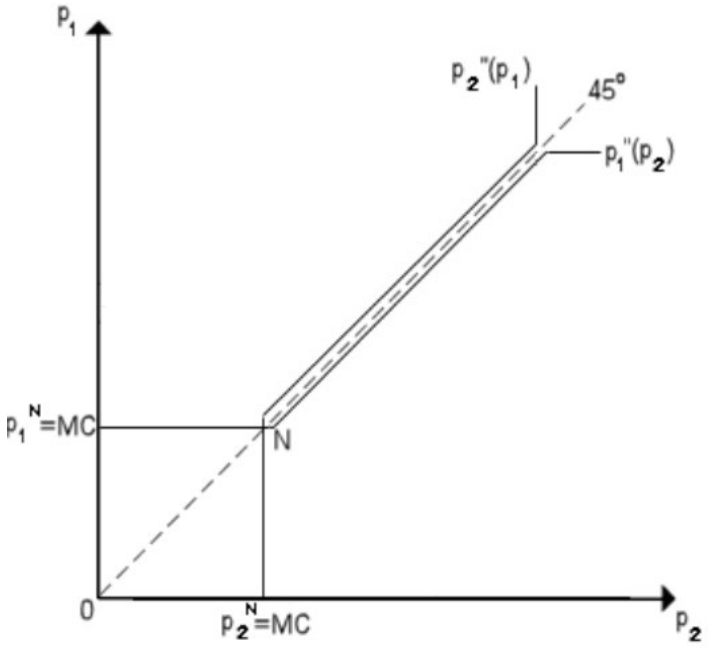
(MC = Marginal costs)

Return = P – MC

If P = MC →

Return = P – MC = 0 →

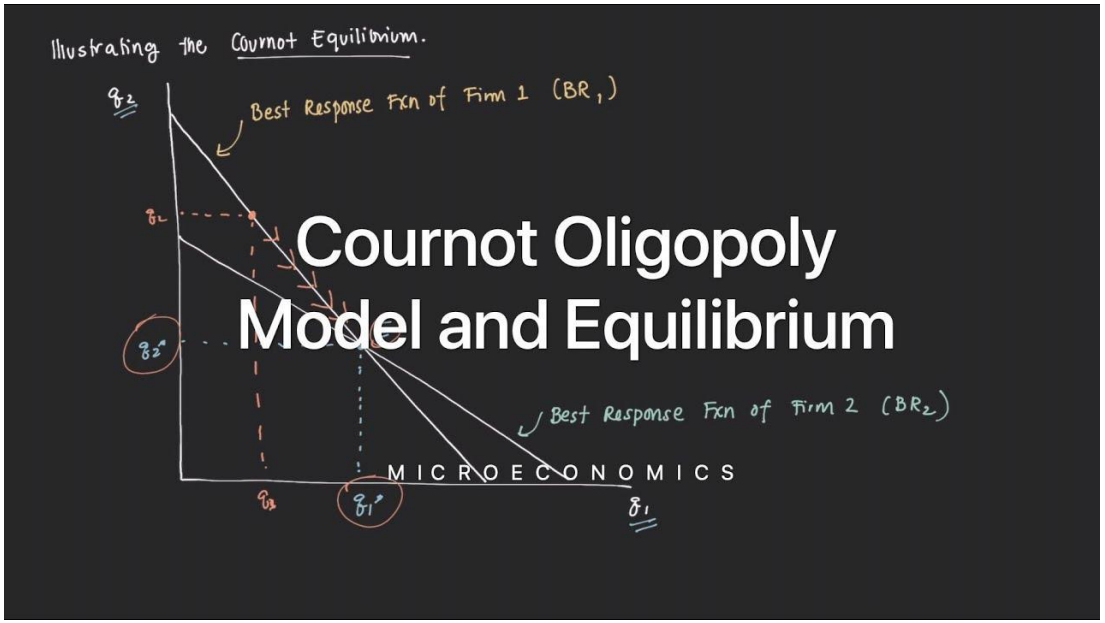
Return is 0 for both duopoly companies if equilibrium occurs.



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Cournot Oligopoly model

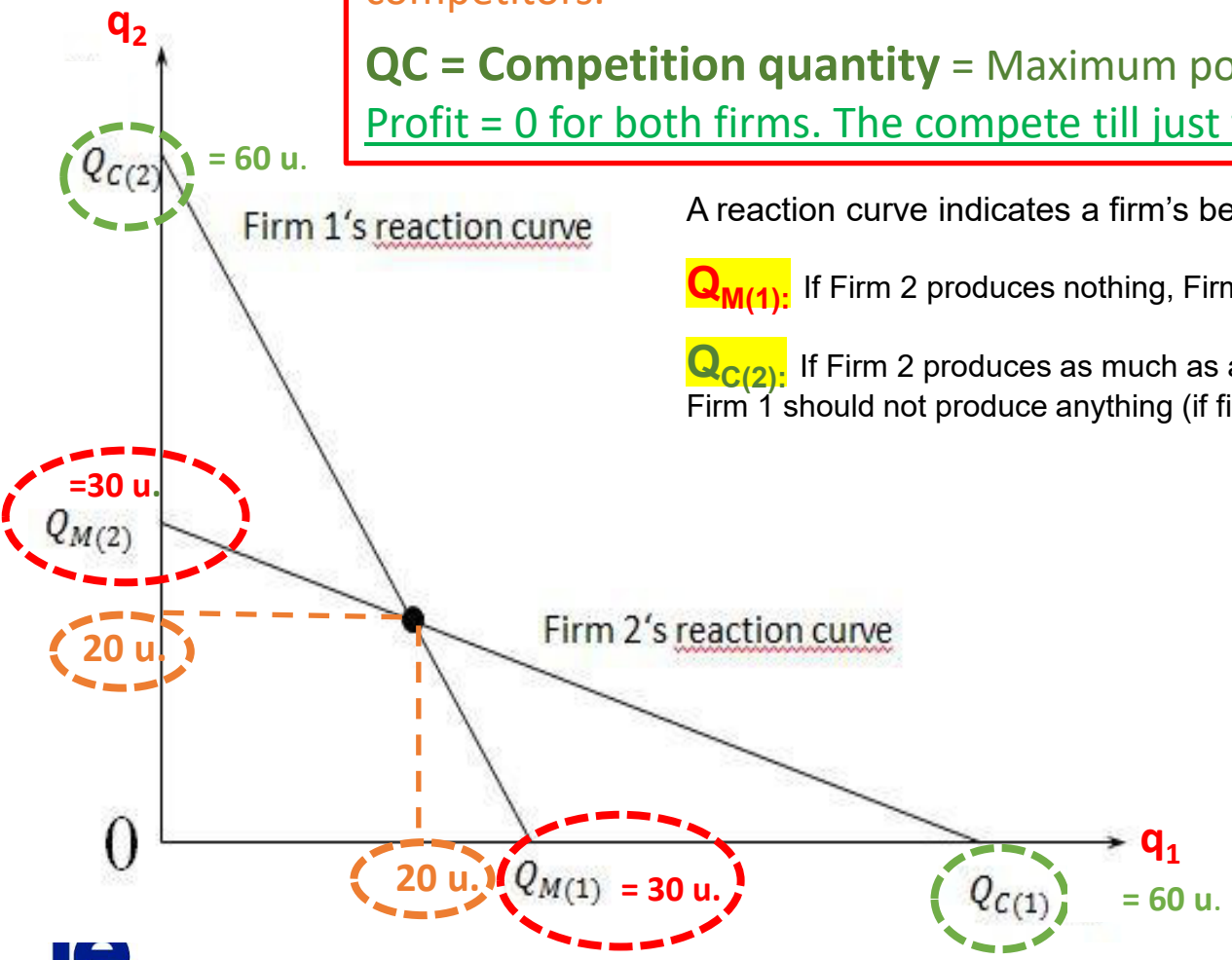
How to Solve a Cournot Oligopoly Problem (9 m) (at home)
https://www.youtube.com/watch?v=An_2r6Ae_28



Q_M = Monopoly quantity = When one of the Firms produce all, in that case they will chose to produce the quantity that maximize their profit.

QD = Cournot quantity = The quantity that makes equal profits for both competitors.

QC = Competition quantity = Maximum possible quantity if they compete. Profit = 0 for both firms. The compete till just they just cover costs.



A reaction curve indicates a firm's best response to the other firm's choice

$Q_{M(1)}$: If Firm 2 produces nothing, Firm 1 will choose the monopoly quantity

$Q_{C(2)}$: If Firm 2 produces as much as an entire competitive industry would, Firm 1 should not produce anything (if fixed costs are not taken into account)

$Q_M = Q_{M1} = Q_{M2}$

$Q_D = Q_{D1} + Q_{D2}$

$Q_C = Q_{C1} = Q_{C2}$

$$Q_D = q_1^* + q_2^*$$

$$Q_M < Q_D < Q_C$$

Monopoly — Cournot — Competition

Cournot model

- *Only two firms in the industry: Firm 1 and Firm 2*
- *Firm1* and *Firm 2, produce identical product and price*
- Strategic Decisions: Produced quantity → q_1, q_2
- *They set the quantity (q_1 & q_2) of production simultaneously.*
- Marginal costs (MC) = c ; are constant
- Profit function of *Firm i:* $\Pi i = qi\,[P_{(q1;q2)} - c]$

Equilibrium price

P → Price

q_1 → quantity Firm 1

q_2 → quantity Firm 2

Q_D → $q_1 + q_2$

c → Marginal costs

Π → Profit

$\Pi_i = q_i\,[P(q1;q2) - c]$

Profit $\Pi_1 = q_1\,[P_{(q1;q2)} - c]$

Profit $\Pi_2 = q_2\,[P_{(q1;q2)} - c]$

The best quantity for Firm 1 depends on quantity of Firm 2

$QM = QM_1 = QM_2$

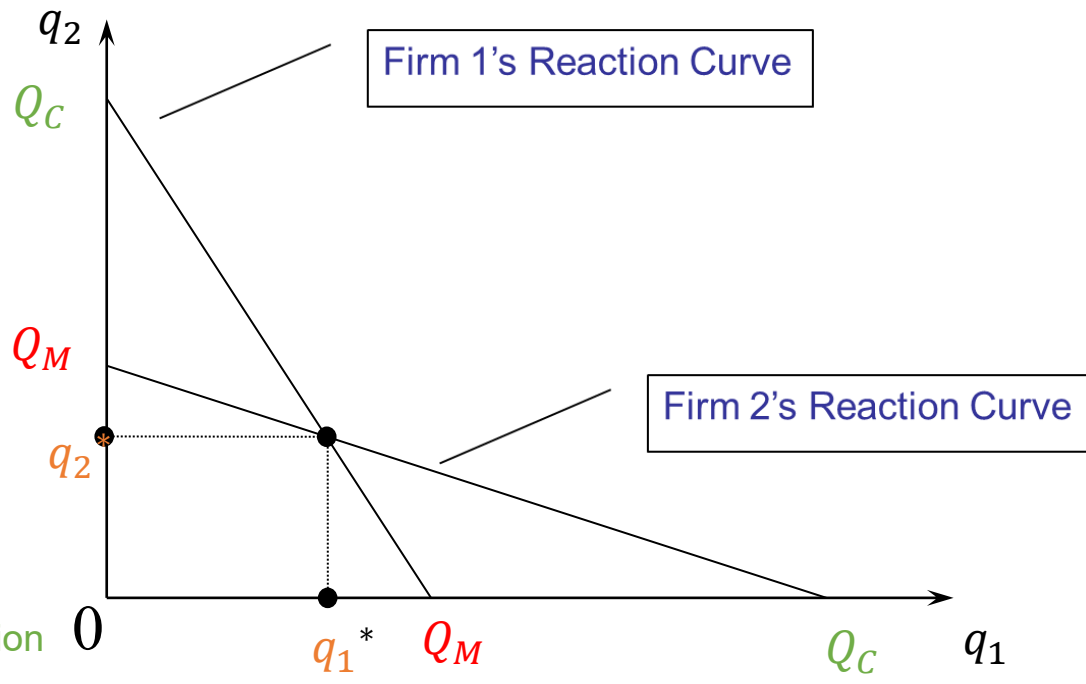
$QD = QD_1 + QD_2$

$QC = QC_1 = QC_2$

$$Q_D = q_1^* + q_2^*$$

$$Q_M < Q_D < Q_C$$

Monopoly — Cournot — Competition



This is what you can find out about the industry

- There are only two firms in this oligopoly (Firm 1 & Firm 2) and the products of both firms are perfect substitutes.
- They sell everything they produce, but it affects the price for both firms. Price is going to be the same for both of them.
- Strategic Decision → **Production quantity.**
- The sales (production) of Firm 1 is q_1 and the sales (production) of Firm 2 is q_2.
- The market demand curve is: $Price = 100 - q_1 - q_2$
- **Profit = Revenue – Total Costs**
- The cost functions are identical for the two firms.

$TC_1 = 40\ q_1$ and $TC_2 = 40\ q_2$ then $TC_{(1+2)} = 40\ (q_1 + q_2)$

What quantity do you recommend Firm 1 and 2 to produce?

We need to know Firm 1's reaction function

- 1st → **Calculate Total Revenue for Firm 1 and 2**
- 2nd → **Calculate Marginal Revenue for Firm 1 and 2**
- 3rd → **Calculate Marginal Cost Firm 1 and 2**
- 4th → **Calculate Number of units at equilibrium for Firm 1 & 2**

Calculate Reaction Function for Firm 1 and 2→

Marginal Revenue Firm 1 = Marginal Total Costs (Firm 1+2)

→ q_1 = function of q_2

Marginal Revenue Firm 2 = Marginal Total Costs (Firm 1+2)

→ q_2 = function of q_1

- 5th → **Obtain equilibrium for Firm 1 and 2 for Production function for maximization profits of both. Calculate PROFIT**

1st → Calculate Total Revenue for Firm 1 and 2

Total Revenue (TR_1)= Price x q_1

$TR_1 = (100 - q_1 - q_2) \times q_1 =$

Price (P) = 100 - q1 - q2

$TR_1 = 100\ q_1 - q_1^2 - q_{1\ x}\ q_2$

Total Revenue (TR_2)= Price x q_2

Price (P) = 100 - q1 - q2

$TR_2 = (100 - q_1 - q_2) \times q_2 =$

$TR_2 = 100\ q_2 - q_2^2 - q_{1\ x}\ q_2$

2nd → Calculate Marginal Revenue for Firm 1 and 2

Marginal revenue (MR) is the increase in revenue that results from the sale of one additional unit of output.

In economic theory, perfectly competitive firms continue producing output until marginal revenue equals marginal cost.

Marginal Revenue = **Derivative (Total Revenue Firm $_1$)**

Calculate the derivative for calculating Marginal Revenue

Marginal Revenue (MR_1) = 100 q_1 – q_1^2 - q_1 x q_2 → ***Derivative***

$MR_1 = 100 - 2q_1 - q_2$

Marginal Revenue (MR_2) = 100 q_2 – q_2^2 - q_1 x q_2 → ***Derivative***

$MR_2 = 100 - 2q_2 - q_1$

3rd → Calculate Marginal Cost Firm 1 and 2

Marginal Costs (MC) is the change in the total cost that arises when the quantity produced is incremented one unit. → **The cost of producing one additional quantity**

In some contexts, it refers to an increment of one unit of output, and in others it refers to the rate of change of total cost as output is increased by an infinitesimal amount.

Marginal Cost = **Derivative (Total Cost Firm $_1$)**

Marginal Cost $_1$ = 40 q_1 → *Derivative* → **Marg Cost $_1$ = 40**

Marginal Cost $_2$ = 40 q_2 → *Derivative* → **Marg Cost $_2$ = 40**

4rd → REACTION FUNCTION

Marginal Revenue = Marginal Costs

$100 - 2q_1 - q_2 = 40 \rightarrow$

$2q_1 = 100 - q_2 - 40 \rightarrow q_1 = (60 - q_2) / 2 \rightarrow \quad q_1 = 30 - 0{,}5\, q_2$

$100 - 2q_2 - q_1 = 40 \rightarrow$

$2q_2 = 100 - q_1 - 40 \rightarrow q_2 = (60 - q_1) / 2 \rightarrow \quad q_2 = 30 - 0{,}5\, q_1$

Replace q2 in the first function and calculate q1

$q_1 = 30 - 0{,}5\, q_2 \rightarrow \quad q_1 = 30 - 0{,}5\,(30 - 0{,}5\, q_1) \rightarrow$

$q_1 = 15 + 0{,}25\, q_1 \rightarrow \quad q_1 = 20$ units

$q_2 = 30 - 0{,}5\, q_1 \rightarrow \quad q_2 = 30 - 0{,}5\,(30 - 0{,}5\, q_2) \rightarrow$

$q_2 = 15 + 0{,}25\, q_2 \rightarrow \quad q_2 = 20$ units

5th → PROFIT CALCULATION

COURNOT → Firm 1 and 2 produce Cournot equilibrium → 20 units

Price (P) = 100 - q1 - q2

Firm 1 = 20 units & Firm 2 = 20 units

Price = 100 -20 -20 = 60 € / unit

TC_1 = 40€ x 20 u. = 800 € **TC_2 = 40€ x 20 u. = 800 €**

$Profit_1 = Profit_2$ = (60€ x 20 u.) – (40€ x 20 u.) = 400 €

5th → PROFIT CALCULATION

MONOPOLY → Firm 1 produce all (30 u.) and Firm 2 nothing

$q_1 = 30 - 0{,}5\ \cancel{q_2} = 30$ u.

$q_2 = 30 - 0{,}5\ \cancel{q_1} = 30$ u.

Firm 1 = 30 units & Firm 2 = 0 units

Price = 100 -30 -0 = 70 € / unit

TC_1 = 40€ x 30 u. = 1200 € **TC_2 = 40€ x 0 u. = 0 €**

$Profit_1$ = (70€ x 30 u.) – (40€ x 30 u.) = 900 €

5th → PROFIT CALCULATION

COMPETITION → Firm 1 and 2 produce → 60 units

Firm 1 = 30 units & Firm 2 = 30 units → 30 + 30 = 60 units

Price = 100 - 30 - 30 = 40 € / unit

0 € = Profit = Revenue – Costs

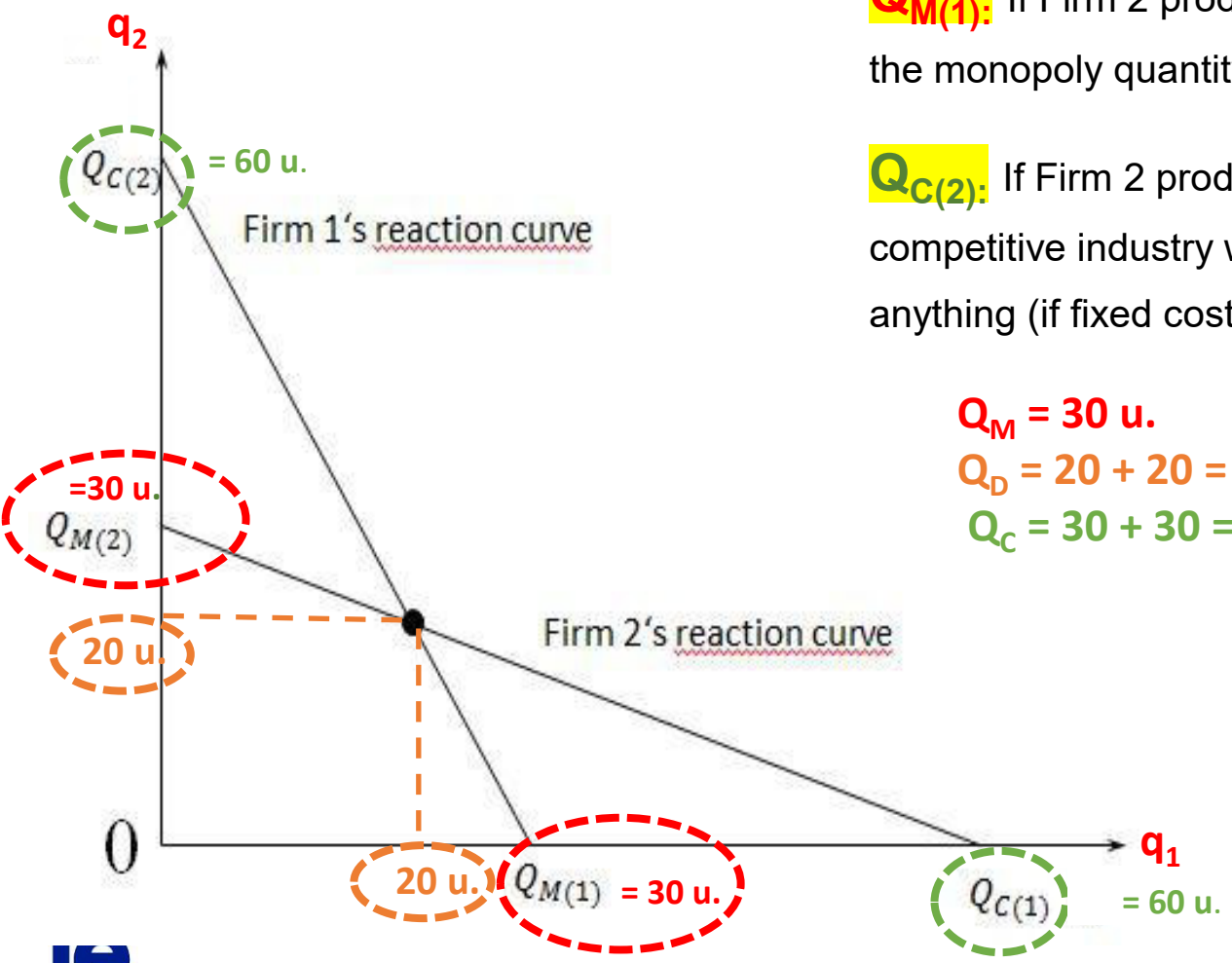
0 € = Profit = (30 units x 40 €/u.) – (30 units x 40 €/u.)

TC_1 = 40€ x 30 u. = 1200 € **TC_2 = 40€ x 30 u. = 1200 €**

$Profit_1$ = 0 → Price = 40€ and 30 units

$q_1 = 30 - 0{,}5\ q_2 = 20$ u.

$q_2 = 30 - 0{,}5\ q_1 = 20$ u.



A reaction curve indicates a firm's best response to the other firm's choice

$Q_{M(1)}$: If Firm 2 produces nothing, Firm 1 will choose the monopoly quantity

$Q_{C(2)}$: If Firm 2 produces as much as an entire competitive industry would, Firm 1 should not produce anything (if fixed costs are not taken into account)

$Q_M = 30$ u.
$Q_D = 20 + 20 = 40$ u.
$Q_C = 30 + 30 = 60$ u.

$Q_M = Q_{M1} = Q_{M2}$
$Q_D = Q_{D1} + Q_{D2}$
$Q_C = Q_{C1} = Q_{C2}$

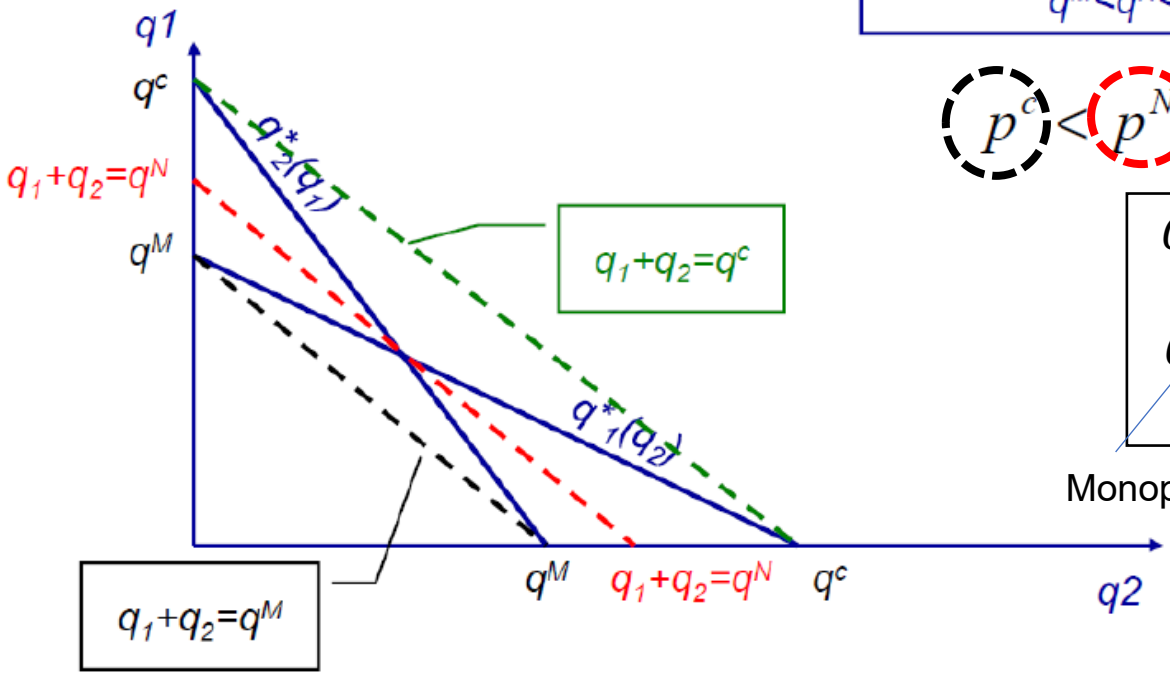
$30 < 40 < 60$

$$Q_D = q_1^* + q_2^*$$

$$Q_M < Q_D < Q_C$$

Monopoly — Cournot — Competition

Comparison between Cournot, Monopoly and Perfect Competition



$q^M < q^N < q^c$

$p^c < p^N < p^M$

$Q_D = q_1^* + q_2^*$

$Q_M < Q_D < Q_C$

Monopoly

Cournot

Competition

Thanks a lot!

Any questions?

rlp@faculty.ie.edu

https://www.youtube.com/watch?v=-RiZiTz2hWA



https://www.youtube.com/watch?v=S49oKtISkqI

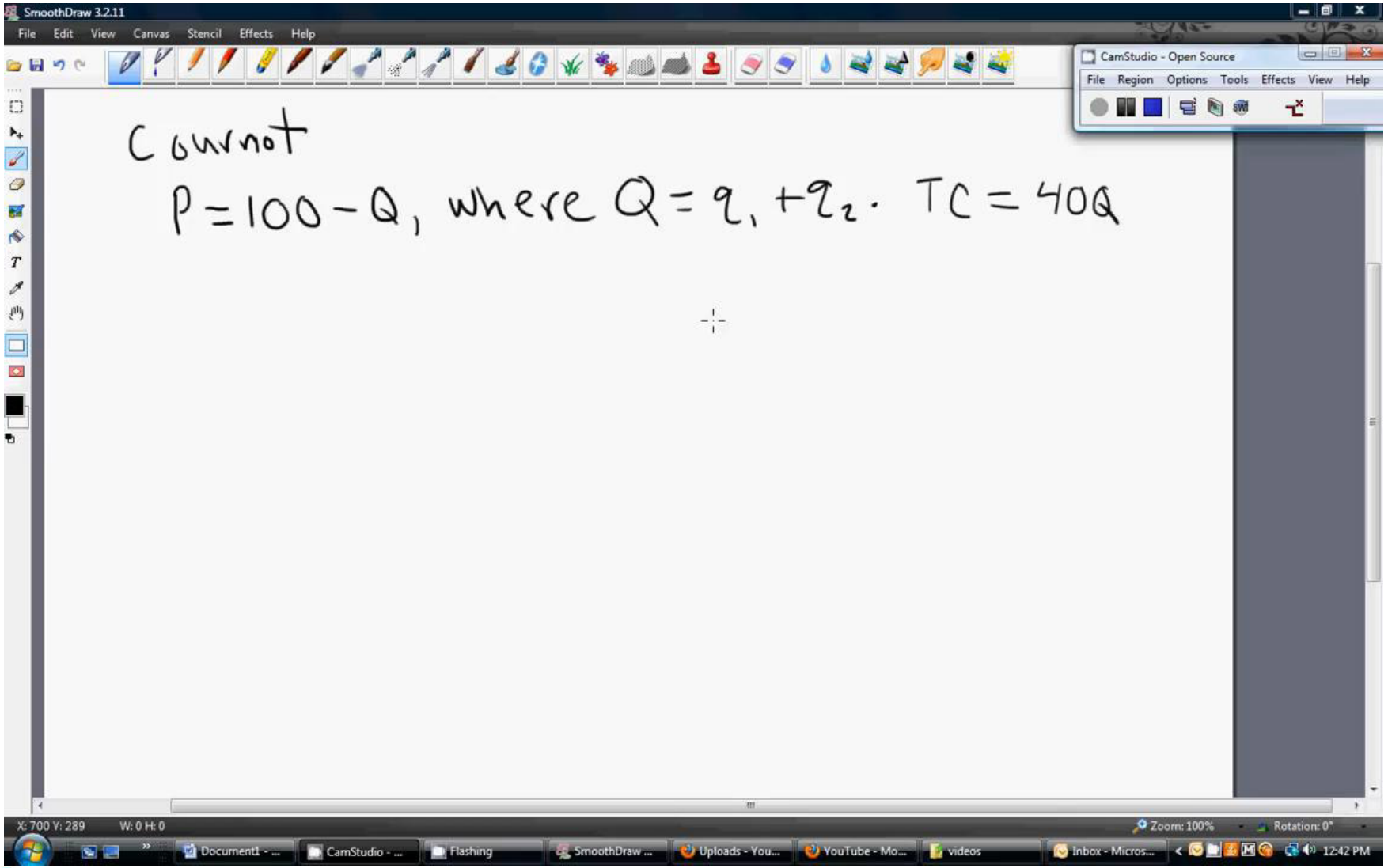


A classroom game: The investment game

- **Players:** All of you
- **Strategies:**
  - Invest 0 Euros
  - Invest 10 Euros
- **Payoffs:**
  - Invest 0 Euros → 0 Euros
  - Invest 10
    - Net profit 5 Euros if at least 90% of the class invests
    - Net loss 10 Euros if less than 90% of the class invests
- **Rules of the Game**:
  - You are not allowed to communicate
  - Write your decision what you do on a little piece of paper
  - Show your neighbor what you did

How to Solve a Cournot Oligopoly Problem (9 m) (at home)

https://www.youtube.com/watch?v=An_2r6Ae_28



Drugmakers Pledge to Avoid Safety Shortcuts on Covid Vaccine



How does this relate to Game theory?

https://www.bloomberg.com/news/articles/2020-09-08/drugmakers-join-to-pledge-no-safety-shortcuts-for-covid-vaccine?utm_campaign=news&utm_medium=bd&utm_source=applenews